LAMMPS diagnostic exclusions

Source-law probes and superseded mappings | 29 October 2026

Five completed diagnostic/superseded executions retain 22,325 native frames. They are excluded from verified comparison bundles and official pass/fail counts. A separate preflight rejection performed zero integration steps.

Three CPU-only Mindlin-history probes

Source-history equivalence FAILS in all three probes. Each source model has tangential ct=0.5cn. The proposed common-Coulomb-saturation reduction fails at an actual saved contact: the source trial can lie inside the disk while the native force is capped. Correct normal forces alone do not establish contact-law equivalence.

CPU probe	Saved frames	First step / time (s)	Source trial / cap (N)	First Ft residual (N)
dem04-oblique-grazing	4863	2237 2.237e-05	0.00453818 0.03025708	0.0257189
dem06-spin-wall	1659	51 5.1e-05	736491.1 353.33	706.6599
dem06-mindlin-spin-heavy	1801	51 5.1e-05	770959.7 337.5601	675.1202

Each residual above exceeds the declared 1e-10 N force-comparison tolerance. These are mapping counterexamples, not failures of the official analytic benchmark. No GPU or half-step validation is claimed for these probes.

Independent all-contact proof

dem04-oblique-grazing: normal-law maximum residual 2.8422e-14 N; tangential-history maximum residual 0.03400539 N; source/native ct at first mismatch 0.06892658/0.1234841 N s/m.

dem06-spin-wall: normal-law maximum residual 1.819e-12 N; tangential-history maximum residual 1810.112 N; source/native ct at first mismatch 2204.615/3949.634 N s/m.

dem06-mindlin-spin-heavy: normal-law maximum residual 3.638e-12 N; tangential-history maximum residual 1745.514 N; source/native ct at first mismatch 2106.375/3773.634 N s/m.

LAMMPS displacement history, damping, total-force Coulomb limit and history back-calculation were replayed without fitting. The source undeformed-radius slip and native shared-point convention are distinguished. Each proof JSON gives the first counterexample, all-contact CSV hash and native input/output hashes.

Pinned LAMMPS source

Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250. Source tags and exact input blobs are preserved in the linked native specifications and proof records. LAMMPS was not executed locally; its published source law was evaluated against actual FunDEM CPU records.

LAMMPS retained diagnostic evidence

Superseded wall modulus mapping

An initial Tsuji sphere-wall mapping incorrectly doubled both material Young moduli. The pinned source and native matching-material expressions both give E_eff=E/[2(1-nu^2)]. The incorrect E-times2 CPU/GPU base runs are preserved as superseded evidence; two fresh corrected base runs and the proper half-step runs form the separate canonical benchmark. A restitution result alone cannot prove force-amplitude equivalence.

The superseded attempt contributes two completed executions and 14,002 frames. Its raw projects, CSV, logs, frame hashes, configuration and source correction record remain retained. It is not an additional official benchmark FAIL.

Cylinder-wall startup rejection

The first prepared-grid attempt failed a native padding preflight before integration. It has no completed simulation, no run.json and zero native simulation frames. The rejection record remains available; the corrected prepared-grid benchmark and its mesh diagnostic are reported separately.

Counting and complete retention

Retained role	Executions	Native frames
dem03-hertz-tsuji-wall-E-times2	2	14,002
dem04-oblique-grazing-cpu-probe	1	4,863
dem06-spin-wall-cpu-probe	1	1,659
dem06-mindlin-spin-heavy-cpu-probe	1	1,801
All excluded completed attempts	5	22,325
Padding startup rejection	0	0

Verified bundles and their primary/mesh-diagnostic totals are listed separately in RESULTS-MANIFEST.json. Grand retained totals add these five excluded executions and 22,325 frames exactly once. Unsupported contact models remain outside verified comparison counts.

Evidence links

dem04-oblique-grazing: proof JSON | all-contact replay CSV | full native ZIP.

dem06-spin-wall: proof JSON | all-contact replay CSV | full native ZIP.

dem06-mindlin-spin-heavy: proof JSON | all-contact replay CSV | full native ZIP.

Excluded-attempt ledger | Modulus correction | Startup rejection.

All five excluded ZIPs pass CRC and native-frame size/SHA checks. Full native results and ZIPs remain in the authoritative Windows delivery folder; compact WSL copies contain inputs, CSV, reports and audits.